

# CSUF Visit Board Notes

## Table of Contents

July 21, 2026

July 22, 2026

July 23, 2026

July 21, 2026

**Plan**

- Solve for  $\vec{u}$  using Poiseuille's pressure sol.
  - error analysis once we have this
- Calculate pressure and compare to true (Poiseuille's) solution
  - plot both
  - analyze error (p will mess this up...)
- What's  $\beta$  doing? → prescribe using outflow? maybe not?  $\epsilon_1$  and  $\epsilon_2$ ?
- Ask Shifra why  $\vec{u} \cdot \hat{n}$  approach to calib.  $\beta$  doesn't work again? Can't we just change perspective?

**Equations:**

$$\frac{dX}{dt} = u(X) + U$$

Circle:  $\vec{u} = 0, U$  doublet =  $\frac{dX}{dt}$

channel:  $\frac{dX}{dt} = 0$

**Diagram:** A schematic of a channel flow. A rectangular channel of height  $2L$  is shown. The left boundary is at  $x = -L$  and the right boundary is at  $x = L$ . The top boundary is at  $y = L$  and the bottom boundary is at  $y = -L$ . The flow is in the  $x$  direction. The velocity profile is shown as a dashed line. The pressure profile is shown as a solid line. The pressure is zero at the inlet and outlet. The pressure is negative in the center of the channel. The velocity is zero at the walls and maximum in the center.

**Equations:**

$$St + SD = U_0 - SD_h$$

$$St + SD + SD_h = U_0$$

$$[p] = - \int_V (\mathbf{h} \cdot \mathbf{n})(\mathbf{n} \cdot \mathbf{x}) Q(\mathbf{r}) d\mathbf{r}$$

**Poiseuille Flow**  $-(L+c) < x < -L$   
 $L < x < L+c$

**Diagram:** A schematic of a channel flow. A rectangular channel of height  $2L$  is shown. The left boundary is at  $x = -L$  and the right boundary is at  $x = L$ . The top boundary is at  $y = L$  and the bottom boundary is at  $y = -L$ . The flow is in the  $x$  direction. The velocity profile is shown as a dashed line. The pressure profile is shown as a solid line. The pressure is zero at the inlet and outlet. The pressure is negative in the center of the channel. The velocity is zero at the walls and maximum in the center.

**Equations:**

$$\nabla p = \mu \Delta \vec{u}$$

$$p(x) = \frac{2p(L)}{\sinh(\alpha L)} \sinh(\alpha x), \quad \alpha^2 = \frac{3\mu}{H^2} Da$$

$$p(x) = \frac{2p(L)}{\sinh(\frac{\sqrt{3\mu Da}}{H^2} L)} \sinh\left(\frac{\sqrt{3\mu Da}}{H^2} x\right)$$

$$\nabla p = \left(\frac{2p(L)}{\sinh(\alpha L)} \cosh(\alpha x), 0\right) = \mu \Delta \vec{u} = (\mu u_{xx}, \mu v_{yy})$$

$$u(x, \pm H) = 0, \quad v(x, \pm H) = 0$$

$$u(-L, y) = 0, \quad v(-L, y) = 0$$

$$u(L, y) = 0, \quad v(L, y) = 0$$

$$u'' + \lambda^2 u = 0, \quad g'' - \lambda^2 g = 0$$

$$f(x) = G \sin(\sqrt{\lambda} x) + G \cos(\sqrt{\lambda} x)$$

$$f(-L) = -G \sin(\sqrt{\lambda} L) + G \cos(\sqrt{\lambda} L) = 0$$

$$f(L) = G \sin(\sqrt{\lambda} L) + G \cos(\sqrt{\lambda} L) = 0$$

$$G \sin(\sqrt{\lambda} L) = G \cos(\sqrt{\lambda} L)$$

$$\tan(\sqrt{\lambda} L) = 1$$

$$\sqrt{\lambda} L = \frac{\pi}{4}$$

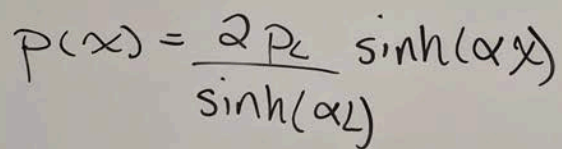
$$\lambda = \left(\frac{\pi}{4L}\right)^2$$

$$G = \frac{p(L)}{\cosh(\alpha L)}$$

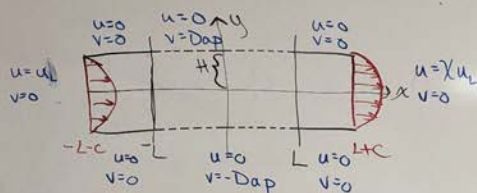
$$p(x) = \frac{p(L)}{\cosh(\alpha L)} \cosh(\alpha x)$$

$$p(L) = \frac{p(L)}{\cosh(\alpha L)} \cosh(\alpha L)$$

$$p(L) = p(L)$$



$$P_x = p'(x)$$



$$p(x) = \frac{2P_0}{\sinh(\alpha L)} \sinh(\alpha x)$$

$$P_x = p'(x)$$

$$B_n = \frac{-\frac{2}{d} \int_0^d f(y) \sin\left(\frac{n\pi y}{d}\right) dy}{\sinh\left(\frac{n\pi b}{d}\right)}$$

$$\nabla p = \mu \Delta \vec{u}$$

$$\vec{u} = (u, v)$$

$$\nabla \cdot \vec{u} = 0$$

Componentwise:

$$\text{horizontal: } P_x = \mu(u_{xx} + u_{yy})$$

$$\text{vertical: } 0 = \mu(v_{xx} + v_{yy})$$

pg. 188 P. Inchenner

Vertical component PDE

$$\mu(v_{xx} + v_{yy}) = 0$$

$$v(-L-c, y) = 0$$

$$v(L+c, y) = 0$$

$$v(x, -H) = h(x) = \begin{cases} 0, & |x| \geq L \\ -Da p(x), & -L < x < L \end{cases}$$

$$v(x, H) = k(x) = \begin{cases} 0, & |x| \geq L \\ Da p(x), & -L < x < L \end{cases}$$

Horizontal component PDE

$$\mu(u_{xx} + u_{yy}) = \frac{2P_0 \alpha}{\sinh(\alpha L)} \cosh(\alpha x)$$

$$u = u_h + u_p$$

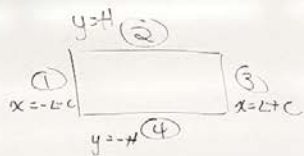
Homogeneous part:

$$u_h(-L-c, y) = f(y) = \frac{Ga^2}{2\mu} \left(1 - \frac{y^2}{H^2}\right)$$

$$u_h(L+c, y) = g(y) = \chi \frac{Ga^2}{2\mu} \left(1 - \frac{y^2}{H^2}\right)$$

$$u_h(x, -H) = 0$$

$$u_h(x, H) = 0$$



$$\Delta u_h = 0$$

$$u_h(-L-c) = u_L(y)$$

$$u_h(L+c) = \chi u_L(y)$$

$$i. \Delta u_h^1 = 0 \quad u_h^1(-L-c, y) = u_L(y) \quad ①$$

$$ii. \Delta u_h^2 = 0 \quad u_h^2(L+c, y) = \chi u_L(y) \quad ③$$

$$u_h^2 = 0 \quad ①, ②, ④$$

$$① \quad u_h^1(-L-c, y) = u_L(y)$$

$$u_h^1(x, H) = 0$$

$$u_h^1(L+c, y) = \chi u_L(y)$$

$$u_h^1(x, -H) = 0$$

$$\Delta u = \frac{C \cosh(\alpha x)}{f(x, y)}$$

$$u = g \text{ on } \partial \Omega$$

$$u = w + v_0$$

$$\Delta v_0 = f \quad \text{up an part soln to } \Delta v_0 = f \quad v_0 = 0 \text{ on } \partial \Omega$$

$$\Delta w = 0 \quad w = g \text{ on } \partial \Omega$$

$$u_H(x,y) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2H} y\right) \left[ A_n \sinh\left(\frac{n\pi}{2H} x\right) + B_n \cosh\left(\frac{n\pi}{2H} x\right) \right]$$

$$\text{BCs: } x = -L-c: u_L = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2H} y\right) \left[ A_n \sinh\left(\frac{n\pi}{2H} (-L-c)\right) + B_n \cosh\left(\frac{n\pi}{2H} (-L-c)\right) \right]$$

$$\frac{GH^2}{2\mu} \left( \frac{1}{H} - \frac{y}{H} \right) u_L = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{2H} (y+H)\right) \quad C_n = \frac{1}{H} \int_{-H}^H \phi(y) \sin\left(\frac{n\pi}{2H} (y+H)\right) dy$$

for  $\phi_1$  &  $\phi_2$

$$\phi(y) = 1: \frac{1}{H} \int_{-H}^H \sin\left(\frac{n\pi}{2H} (y+H)\right) dy$$

$$= -\cos\left(\frac{n\pi}{2H} (y+H)\right) \frac{2}{n\pi} \Big|_{-H}^H$$

$$= -\cos(n\pi) - (-\cos(0)) = -\cos(n\pi) + 1$$

$$= \begin{cases} 0, & n \text{ even} \\ 2, & n \text{ odd} \end{cases} \quad C_n = \begin{cases} 0 & n \text{ even} \\ 2 & n \text{ odd} \end{cases}$$

Cases:  $\lambda = 0 \Rightarrow Y = c_1 + c_2 y$

$$Y(H) = c_1 + H c_2 = 0$$

$$Y(-H) = c_1 - H c_2 = 0$$

$$\Rightarrow c_1 = c_2 = 0$$

Cases:  $\lambda < 0 \Rightarrow Y'' - \lambda Y = 0 \Rightarrow Y = c_1 e^{\sqrt{\lambda} y} + c_2 e^{-\sqrt{\lambda} y}$

$$Y(H) = c_1 e^{\sqrt{\lambda} H} + c_2 e^{-\sqrt{\lambda} H} = 0$$

$$Y(-H) = c_1 e^{-\sqrt{\lambda} H} + c_2 e^{\sqrt{\lambda} H} = 0$$

$$2c_1 \cosh(\sqrt{\lambda} H) + 2c_2 \cosh(\sqrt{\lambda} H) = 0$$

$$(2c_1 + 2c_2) \cosh(\sqrt{\lambda} H) = 0$$

$$c_1 + c_2 = 0 \rightarrow \text{no sol.}$$

$$c_1 - c_2 = 0$$

Cases:  $\lambda > 0$

$$\lambda = \left(\frac{n\pi}{2H}\right)^2$$

$$\sin(n\pi) = 0$$

$$Y'' + \lambda Y = 0 \quad Y(H) = 0 \quad Y(-H) = 0$$

$$\cos(\sqrt{\lambda} H) = 0$$

$$\sqrt{\lambda} = \frac{(2n+1)\pi}{2H}$$

$$\lambda = \left(\frac{(2n+1)\pi}{2H}\right)^2$$

$y \rightarrow y+H$

$$Y'' + \lambda Y = 0 \quad Y(0) = 0, Y(2H) = 0$$

$$\lambda > 0 \quad Y(y) = c_1 \sin(\sqrt{\lambda} y) + c_2 \cos(\sqrt{\lambda} y)$$

$$Y(0) = c_2 = 0$$

$$Y(2H) = c_1 \sin(\sqrt{\lambda} 2H) = 0$$

$$\sqrt{\lambda} 2H = n\pi \quad \lambda = \left(\frac{n\pi}{2H}\right)^2$$

$$Y(y) = c_1 \sin\left(\frac{n\pi}{2H} y\right)$$

$$\phi(y) = y^2: C_n = \frac{1}{H} \int_{-H}^H y^2 \sin\left(\frac{n\pi}{2H} (y+H)\right) dy = \frac{-(\pi^2 n^2 - 8)(\cos(n\pi) - 1)(2H^2)}{n^3 \pi^3} = \begin{cases} 0, & n \text{ even} \\ \frac{4H^2(\pi^2 n^2 - 8)}{n^3 \pi^3}, & n \text{ odd} \end{cases}$$

$$\tilde{y} = y+H$$

$$= \frac{1}{4} \int_0^{2H} (\tilde{y}-H)^2 \sin\left(\frac{n\pi}{2H} \tilde{y}\right) d\tilde{y}$$

$$= \frac{1}{H} \int_0^{2H} \tilde{y}^2 \sin(K \tilde{y}) d\tilde{y} - 2 \int_0^{2H} \tilde{y} \sin(K \tilde{y}) d\tilde{y} + H \int_0^{2H} \sin(K \tilde{y}) d\tilde{y}$$

$$= -\frac{8H^2}{\pi^3 n^3} (\pi^2 n^2 - 2) \cos(n\pi) + 2 + 8 \left( \frac{H^2 \cos(n\pi)}{n\pi} \right) - \frac{2H}{n\pi} (-\cos(n\pi) + 1)$$

$$= -8H^2 \left( \frac{1}{n\pi} \quad n \text{ even} \right) + \frac{8H^2}{n\pi} (-1)^n - \frac{2H}{n\pi} \begin{pmatrix} 0 & n \text{ even} \\ 2 & n \text{ odd} \end{pmatrix}$$



$$\frac{GH^2}{2\mu} \left(1 - \frac{y^2}{H^2}\right) = u_L = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{2H}(y+H)\right) = \sum_{n=1}^{\infty} \frac{GH^2}{2\mu} \left(2 - \frac{4(\pi^2(an-b^2-8))}{(2n-1)^3\pi^3}\right) \sin\left(\frac{(2n-1)\pi}{2H}(y+H)\right)$$

$$C_n = \begin{cases} 0, & n \text{ even} \\ \frac{GH^2}{2\mu} \left[2 - \frac{4(\pi^2n^2-8)}{n^3\pi^3}\right], & n \text{ odd} \end{cases}$$



$$\frac{GH^2}{2\mu} \left(1 - \frac{y^2}{H^2}\right) = u_L = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{2H}(y+H)\right) = \sum_{n=1}^{\infty} \frac{GH^2}{2\mu} \left(2 - \frac{4(\pi^2(2n-1)^2 - 8)}{(2n-1)^3 \pi^3}\right) \sin\left(\frac{(2n-1)\pi}{2H}(y+H)\right)$$

$$C_n = \begin{cases} 0, & n \text{ even} \\ \frac{GH^2}{2\mu} \left[2 - \frac{4(\pi^2 n^2 - 8)}{n^3 \pi^3}\right], & n \text{ odd} \end{cases}$$

$$\textcircled{1} \Delta u = 0 \quad \begin{aligned} u(-L-c, y) &= u_L \\ u(L+c, y) &= \chi u_L \\ u(x, -H) &= 0 \\ u(x, H) &= 0 \end{aligned}$$

$$\textcircled{3} Y'' + \lambda Y = 0 \quad \lambda > 0 \text{ only eigenvalue}$$

$$\begin{aligned} Y_L(y) &= \sin\left(\frac{n\pi}{2H}(y+H)\right) \\ Y_L(H) &= 0 \quad \checkmark \quad \lambda^2 = \left(\frac{n\pi}{2H}\right)^2 \\ Y_L(-H) &= 0 \end{aligned}$$

$$\textcircled{2} u = X(x)Y(y) \quad \frac{X''}{X} = -\frac{Y''}{Y} = \lambda$$

$$X''Y + Y''X = 0$$

$$\textcircled{4} X'' - \lambda X = 0$$

$$X_n(x) = A_n \sinh\left(\frac{n\pi}{2H}x\right) + B_n \cosh\left(\frac{n\pi}{2H}x\right)$$

$$u(x, y) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2H}(y+H)\right) \left\{ A_n \sinh\left(\frac{n\pi}{2H}x\right) + B_n \cosh\left(\frac{n\pi}{2H}x\right) \right\}$$

$$u_L = u(-L-c, y) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2H}(y+H)\right) \underbrace{\left\{ A_n \sinh\left(\frac{n\pi}{2H}(-L-c)\right) + B_n \cosh\left(\frac{n\pi}{2H}(-L-c)\right) \right\}}_{C_n}$$

$$\chi u_L = u(L+c, y) =$$

$$C_n = -A_n \sinh\left(\frac{\pi}{2}\right) + B_n \cosh\left(\frac{\pi}{2}\right)$$

$$\tilde{C}_n = A_n \sinh\left(\frac{\pi}{2}\right) + B_n \cosh\left(\frac{\pi}{2}\right)$$

$$C_n + \tilde{C}_n = 2 B_n \cosh\left(\frac{\pi}{2}\right)$$

$$\Rightarrow B_n = C_n / \cosh\left(\frac{\pi}{2}\right)$$

$$C_n - \tilde{C}_n = -2 A_n \sinh\left(\frac{\pi}{2}\right)$$

July 22, 2026

$u=0$  at  $y=H$   
 $v=0$  at  $x=-L-c$  and  $x=L+c$   
 $u=0$  at  $x=-L-c$  and  $x=L+c$   
 $v=0$  at  $y=H$  and  $y=-H$

$p(x) = \frac{2P}{\sinh(\alpha L)} \sinh(\alpha x)$   
 $P_x = p'(x)$

$B_n = -\frac{2 \int_0^d f(y) \sin(\frac{n\pi y}{d}) dy}{\sinh(\frac{n\pi b}{d})}$

$\nabla p = \mu \Delta \vec{u}$   
 $\vec{u} = (u, v)$   
 $\nabla \cdot \vec{u} = 0$

Componentwise:

Horizontal:  $P_x = \mu(u_{xx} + v_{yy})$

Vertical:  $0 = \mu(v_{xx} + u_{yy})$

Plan:

- Reformulate to start w/  $[0, L+2c]$
- permeable region  $[c, L+c]$
- $L_y [0, 2H]$
- Use Shilpa's book w/  $b=L+2c$   
 $d=2H$

PS 188 Qindhuur  
 Vertical compo  
 $\mu(v_{xx} + u_{yy})$   
 $v(-L-c, y)$   
 $v(L+c, y)$   
 $v(x, -H)$   
 $v(x, H)$

wise:

$\mu(u_{xx} + v_{yy})$   
 $\mu(v_{xx} + u_{yy})$

Plan:

- Reformulate to start w/  $[0, L+2c]$
- permeable region  $[c, L+c]$
- $L_y [0, 2H]$
- Use Shilpa's book w/  $b=L+2c$   
 $d=2H$

$v(-L-c, y) = 0$   
 $v(L+c, y) = 0$   
 $v(x, -H) = h(x) = \begin{cases} 0, & |x| \geq L \\ -Da p(x), & -L < x < L \end{cases}$   
 $v(x, H) = k(x) = \begin{cases} 0, & |x| \geq L \\ Da p(x), & -L < x < L \end{cases}$

- Interpret  $P_L$  for 2D
- Conservation of mass ( $\chi$  and  $\beta$  relationship)
- validity wrt boundary layer & radius of channel (8 m cp)

$2H$   
 $0$   $c$   $L+c$   $L+2c$

$\mu \Delta u = P_x$   
 $u = u_h + u_p$

$u_h$  solves:  
 $\mu \Delta u = 0$   
 $u(0, y) = u_L$   
 $u(L+2c, y) = \chi u_L$   
 $u(x, 0) = u(x, 2H) = 0$

We have:  $v(x, 0) = -Da P$   
 $v(x, 2H) = Da P$

Control volume:  $Q_x = -2Da P$

$u_p$  solves:  
 $\mu \Delta u = P_x$   
 $u(0, y) - u(L+2c, y) = u(x, 0) = u(x, 2H) = 0$

What is  $P_x$ ?

$u_L = -\frac{GH}{\mu} y \left( \frac{y}{2H} - 1 \right)$   
 $Q = \int_0^{2H} u_L dy = \int_0^{2H} \frac{GH}{\mu} y \left( \frac{y}{2H} - 1 \right) dy = \frac{GH}{\mu} \left( \frac{1}{2} y^2 - \frac{1}{6H} y^3 \right) \Big|_0^{2H} = \frac{GH}{\mu} \left( 2H^2 - \frac{8H^3}{6H} \right)$

$Q_x = -\frac{GH}{\mu} \left( \frac{2}{3} H^2 \right) = -\frac{2GH^3}{3\mu} = -\frac{2P_x H^3}{3\mu}$   
 $Q_x = -\frac{2H^3}{3\mu} P_{xx} = -2Da P$  (ODE)

not constant in channel  
 $\downarrow$



$$0 = \int_0^{2H} u(x+dx, y) - u(x, y) dy + (v(x, 2H) - v(x, 0)) dx$$

$$0 = \frac{Q(x+dx) - Q(x)}{dx} + v(x, 2H) - v(x, 0)$$

$$\downarrow dx \rightarrow 0$$

$$0 = Q_x + 2DaP$$

$$P_x = \frac{3\mu}{H^3} DaP$$

$$p(x) = c_1 \cosh(\alpha x) + c_2 \sinh(\alpha x)$$

$$= c_1 \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) + c_2 \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$



$$p(c) = -p(L+c)$$

$$p(c) = c_1 \cosh\left(\alpha\left(\frac{L}{2}\right)\right) + c_2 \sinh\left(\alpha\left(\frac{L}{2}\right)\right)$$

$$+ p(L+c) = c_1 \cosh\left(\alpha\left(\frac{L}{2}\right)\right) + c_2 \sinh\left(\alpha\left(\frac{L}{2}\right)\right)$$

$$0 = 2c_1 \cosh\left(\alpha\left(\frac{L}{2}\right)\right)$$

$$c_1 = 0$$

$$p(x) = c_2 \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \frac{p(L+c)}{\sinh\left(\alpha\left(\frac{L}{2}\right)\right)} \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \gamma \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

for  $c_2$ , evaluate at  $x=L+c$ .

$$p(L+c) = c_2 \sinh\left(\alpha\left(\frac{L}{2}\right)\right)$$

$$c_2 = \frac{p(L+c)}{\sinh\left(\alpha\left(\frac{L}{2}\right)\right)}$$

$$u(0, y) = 0$$

$$u(L+2c, y) = 0$$

$$u(x, 0) = 0$$

$$u(x, 2H) = 0$$

$$\Rightarrow p_x(x) = \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

$$u_{xx} + u_{yy} = \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

rhs of  
up eq.

$$u_{xx} = \gamma \alpha \cosh(\dots)$$

$$u_{yy} = \frac{\gamma}{2} \cosh(\dots) - \frac{\gamma}{2} \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right)$$

$$u(0) = \frac{\gamma}{2} \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right) \neq u(L+2c)$$

$$\gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \sum_{n=1}^{\infty} F_n \sin\left(\frac{n\pi x}{L+2c}\right)$$

$$F_n = \frac{2}{L+2c} \int_0^{L+2c} \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) \sin\left(\frac{n\pi x}{L+2c}\right) dx$$

$$= \frac{2}{L+2c} \gamma \alpha \left( \frac{(L+2c)/2 \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right)}{\alpha^2 (L+2c)^2 + (\pi n)^2} \right)$$

$$F_n = \frac{4n\pi \gamma \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right)}{\alpha (L+2c)^2 + (\pi n)^2}$$



$$u(x,y) = \sum_{n=1}^{\infty} A_n(y) \sin\left(\frac{n\pi x}{L+2c}\right)$$

$$u_{xx} + u_{yy} = \sum F_n \sin\left(\frac{n\pi x}{L+2c}\right)$$

$$\sum \left( A_n''(y) \left(\frac{n\pi}{L+2c}\right)^2 \sin\left(\frac{n\pi x}{L+2c}\right) + A_n''(y) \sin\left(\frac{n\pi x}{L+2c}\right) \right) = \sum F_n \sin\left(\frac{n\pi x}{L+2c}\right)$$

$$A_n''(y) - \left(\frac{n\pi}{L+2c}\right)^2 A_n(y) = F_n$$

$$A_n(y) = \underbrace{C_1 \sinh\left(\frac{n\pi}{L+2c} y\right) + C_2 \cosh\left(\frac{n\pi}{L+2c} y\right)}_{A_n^{\text{hom}}} + A_n^p(y)$$

$$A_n^p(y) = c$$

plug in  
trial sol.

$$0 - \left(\frac{n\pi}{L+2c}\right)^2 c = F_n$$

$$c = -\frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2}$$

$$A_n(y) = C_1 \sinh\left(\frac{n\pi}{L+2c} y\right) + C_2 \cosh\left(\frac{n\pi}{L+2c} y\right) - \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2}$$

$$u(x,y) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L+2c}\right) \left[ C_n \sinh\left(\frac{n\pi}{L+2c} y\right) + d_n \cosh\left(\frac{n\pi}{L+2c} y\right) - \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2} \right]$$

BCs:  $u(x,0) = 0 \rightarrow \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L+2c}\right) \left[ d_n - \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2} \right] = 0 \rightarrow d_n = \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2}$

$$u(x,2H) = 0 \rightarrow \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L+2c}\right) \left[ C_n \sinh\left(\frac{2n\pi H}{L+2c}\right) + \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2} \cosh\left(\frac{2n\pi H}{L+2c}\right) - \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2} \right]$$

$$C_n = \frac{1}{\sinh\left(\frac{2n\pi H}{L+2c}\right)} \cdot \frac{F_n}{\left(\frac{n\pi}{L+2c}\right)^2} \left( 1 - \cosh\left(\frac{2n\pi H}{L+2c}\right) \right)$$

$$p(x) = c_2 \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \underbrace{\frac{p(L+c)}{\sinh\left(\alpha\frac{L}{2}\right)}}_{\gamma} \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \gamma \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

for  $c_1$ , evaluate at  $x=L+c$ :

$$p(L+c) = c_2 \sinh\left(\alpha\frac{L}{2}\right)$$

$$c_2 = \frac{p(L+c)}{\sinh\left(\alpha\frac{L}{2}\right)} = \gamma$$

$$\Rightarrow p_x(x) = \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

$$u_{xx} + u_{yy} = \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

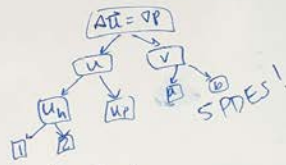
$$\begin{aligned} u(0,y) &= 0 \\ u(L+2c,y) &= 0 \\ u(x,0) &= 0 \\ u(x,2H) &= 0 \end{aligned}$$

rhs of  
up eq.

$$u_{xx} = \gamma \alpha \cosh(\dots)$$

$$u(x) = \frac{\gamma}{\alpha^2} \cosh(\dots) - \frac{\gamma}{2} \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right)$$

$$u(0) = \frac{\gamma}{\alpha^2} \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right) \neq u(L+2c)$$



Discontinuous

$$p(c) = \gamma \sinh\left(\alpha\frac{L}{2}\right) = -p(L+c)$$

$$\gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) = \sum_{n=1}^{\infty} F_n \sin\left(\frac{n\pi x}{L+2c}\right)$$

$$F_n = \frac{2}{L+2c} \int_0^{L+2c} \gamma \alpha \cosh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right) \sin\left(\frac{n\pi x}{L+2c}\right) dx$$

$$= \frac{2}{L+2c} \gamma \alpha \left( \frac{(L+2c) (2n\pi \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right) \sin^2\left(\frac{n\pi}{2}\right))}{\alpha^2 (L+2c)^2 + (\pi n)^2} \right)$$

$$F_n = \frac{4 \pi \gamma \cosh\left(\alpha\left(\frac{L+2c}{2}\right)\right) (2n-1)}{\alpha^2 (L+2c)^2 + (\pi (2n-1))^2}$$

July 23, 2026

$$v_{xx} + v_{yy} = 0$$

$$v(0, y) = v(L+2c, y) = 0$$

$$v(x, 0) = h(x)$$

$$v(x, 2H) = k(x)$$

$$h(x) = \begin{cases} 0 & 0 \leq x < c, L+c \leq x < L+2c \\ -Da\rho & c \leq x < L+c \end{cases}$$

$$k(x) = \begin{cases} 0 & 0 \leq x < c, L+c \leq x < L+2c \\ Da\rho & c \leq x < L+c \end{cases}$$

$$p(x) = \delta \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

$$\delta = \frac{p(L+c)}{\sinh(\alpha L/2)}$$

$$X''Y + XY'' = 0$$

$$X'' + \lambda X = 0 \quad X(0) = 0 \quad X(L+2c) = 0$$

( $\lambda < 0$  trivial)

$$\lambda > 0$$

$$X(x) = c_1 \sin(\sqrt{\lambda}x) + c_2 \cos(\sqrt{\lambda}x)$$

$$X(0) = 0 \Rightarrow c_2 = 0$$

$$X(L+2c) = 0 \quad 0 = c_1 \sin(\sqrt{\lambda}(L+2c))$$

$$\lambda = \left(\frac{n\pi}{L+2c}\right)^2 \quad \sqrt{\lambda} = \frac{n\pi}{L+2c} = n\pi$$

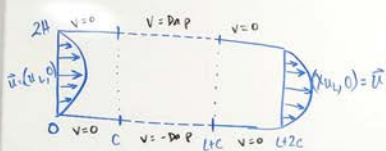
$$X_n(x) = \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$Y'' - \lambda Y = 0$$

$$Y(y) = A_n \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n \cosh\left(\frac{n\pi}{L+2c}y\right)$$

$$-\frac{X''}{X} = \frac{Y''}{Y} = \lambda$$

$$\vec{u} = (u, v)$$



$$\text{B.C.: } v(x, 0) = h(x)$$

$$v(x, y) = \sum_{n=1}^{\infty} \left( A_n \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n \cosh\left(\frac{n\pi}{L+2c}y\right) \right) \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$v(x, 0) = h(x)$$

$$h(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$k(x) = \sum_{n=1}^{\infty} \left( A_n \sinh\left(\frac{n\pi}{L+2c}2H\right) + B_n \cosh\left(\frac{n\pi}{L+2c}2H\right) \right) \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$B_n = \frac{2}{L+2c} \int_0^{L+2c} h(x) \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$= -\frac{2}{L+2c} \int_c^{L+c} Da\rho \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$k(x): \quad Y_n^{(1)}(y) = A_n^{(1)} \sinh\left(\frac{n\pi}{L+2c}(2H-y)\right) + B_n^{(1)} \cosh\left(\frac{n\pi}{L+2c}(2H-y)\right)$$

$$h(x): \quad Y_n^{(2)}(y) = A_n^{(2)} \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n^{(2)} \cosh\left(\frac{n\pi}{L+2c}y\right)$$

$$h(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$B_n^{(2)} = \frac{2}{L+2c} \int_c^{L+c} p(x) \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$B_n = \left( \frac{2\delta Da}{\alpha^2(L+2c)^2 \sinh(\alpha L/2)} \right) \left[ n\pi \sinh\left(\frac{\alpha L}{2}\right) \cos\left(\frac{n\pi c}{L+2c}\right) + \right.$$

$$(1) \quad k(x) = v(x, 2H) = \sum_{n=1}^{\infty} B_n^{(1)} \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$B_n^{(1)} = -B_n^{(2)}$$

$$\cos\left(\frac{n\pi(L+c)}{L+2c}\right) + \alpha(L+2c) \cosh\left(\frac{\alpha L}{2}\right) \left( \sin\left(\frac{n\pi c}{L+2c}\right) - \sin\left(\frac{n\pi(L+c)}{L+2c}\right) \right)$$



$$v_{xx} + v_{yy} = 0$$

$$v(0, y) = v(L+2c, y) = 0$$

$$v(x, 0) = h(x)$$

$$v(x, 2H) = k(x)$$

$$h(x) = \begin{cases} 0 & 0 < x < c, L+c < x < L+2c \\ -2ap & c < x < L+c \end{cases}$$

$$k(x) = \begin{cases} 0 & 0 < x < c, L+c < x < L+2c \\ 2ap & c < x < L+c \end{cases}$$

$$p(x) = \delta \sinh\left(\alpha\left(x - \frac{L+2c}{2}\right)\right)$$

$$\delta = \frac{p(L+c)}{\sinh(\alpha L/2)}$$

$$X''Y + XY'' = 0$$

$$X'' + \lambda X = 0 \quad X(0) = 0 \quad X(L+2c) = 0$$

$$(\lambda < 0 \text{ trivial})$$

$$\lambda > 0$$

$$X(x) = c_1 \sin(\sqrt{\lambda}x) + c_2 \cos(\sqrt{\lambda}x)$$

$$X(0) = 0 \Rightarrow c_2 = 0$$

$$X(L+2c) = 0 \Rightarrow 0 = c_1 \sin(\sqrt{\lambda}(L+2c))$$

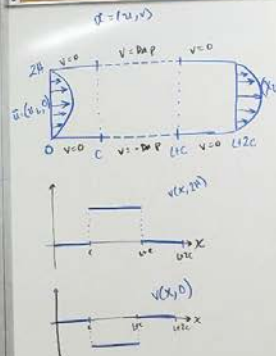
$$\lambda = \left(\frac{n\pi}{L+2c}\right)^2 \quad \sqrt{\lambda} = \frac{n\pi}{L+2c} = n\pi$$

$$X_n(x) = \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$Y'' - \lambda Y = 0$$

$$Y(y) = A_n \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n \cosh\left(\frac{n\pi}{L+2c}y\right)$$

$$-\frac{X'}{X} = \frac{Y''}{Y} = \lambda$$



$$X(0) = 0 \quad X(L+2c) = 0$$

$$-\frac{X'}{X} = \frac{Y''}{Y} = \lambda$$

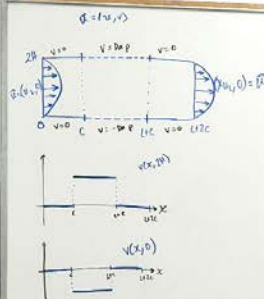
$$c_1 \sin(\sqrt{\lambda}x) + c_2 \cos(\sqrt{\lambda}x)$$

$$c_2 = 0 \quad 0 = c_1 \sin(\sqrt{\lambda}(L+2c))$$

$$\sqrt{\lambda} = \frac{n\pi}{L+2c} = n\pi$$

$$\sin\left(\frac{n\pi}{L+2c}x\right)$$

$$A_n \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n \cosh\left(\frac{n\pi}{L+2c}y\right)$$



$$v(x, y) = h(x)$$

$$v(x, y) = \sum_{n=1}^{\infty} \left( A_n \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n \cosh\left(\frac{n\pi}{L+2c}y\right) \right) \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$v(x, 0) = h(x)$$

$$h(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$k(x) = \sum_{n=1}^{\infty} \left( A_n \sinh\left(\frac{n\pi}{L+2c}2H\right) + B_n \cosh\left(\frac{n\pi}{L+2c}2H\right) \right) \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$B_n = \frac{2}{L+2c} \int_0^{L+2c} h(x) \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$= \frac{2}{L+2c} \int_c^{L+c} 2ap \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$k(x): Y_n^{(1)}(y) = A_n^{(1)} \sinh\left(\frac{n\pi}{L+2c}(2H-y)\right) + B_n^{(1)} \cosh\left(\frac{n\pi}{L+2c}(2H-y)\right)$$

$$h(x): Y_n^{(2)}(y) = A_n^{(2)} \sinh\left(\frac{n\pi}{L+2c}y\right) + B_n^{(2)} \cosh\left(\frac{n\pi}{L+2c}y\right)$$

$$h(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L+2c}x\right)$$

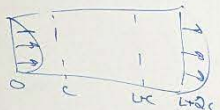
$$B_n^{(2)} = \frac{2}{L+2c} \int_0^{L+2c} p(x) \sin\left(\frac{n\pi}{L+2c}x\right) dx$$

$$B_n^{(2)} = \left( \frac{2\chi Da}{\alpha^2(L+2c)^2(n\pi)^2} \right) \left[ n\pi \sinh\left(\frac{\alpha L}{2}\right) \cos\left(\frac{n\pi c}{L+2c}\right) + \right.$$

$$(1) \quad K(\infty) = V(x, 2H) = \sum_{n=1}^{\infty} k_n^{(1)} \sin\left(\frac{n\pi}{L+2c}x\right)$$

$$B_n^{(1)} = -B_n^{(2)}$$

$$\left. \cos\left(\frac{n\pi(L+c)}{L+2c}\right) + \alpha(L+2c) \cosh\left(\frac{\alpha L}{2}\right) \left( \sin\left(\frac{n\pi c}{L+2c}\right) - \sin\left(\frac{n\pi(L+c)}{L+2c}\right) \right) \right]$$



Finding  $p(c)$  &  $p(L+c)$

(Recall  $p(c) = -p(L+c)$ )  
(Only true when  $\chi = 1$ )

$$-G \equiv p_x(c, y)$$

$$p_x(c, y) = \chi \alpha \cosh\left(\alpha\left(c - \frac{L+2c}{2}\right)\right)$$

$$= \chi \alpha \cosh\left(-\frac{\alpha L}{2}\right)$$

$$= \chi \alpha \cosh\left(\frac{\alpha L}{2}\right)$$

$$-G = \chi \alpha \cosh\left(\frac{\alpha L}{2}\right)$$

$$\frac{p(L+c)}{\sinh\left(\frac{\alpha L}{2}\right)} = \frac{-G}{\alpha \cosh\left(\frac{\alpha L}{2}\right)}$$

$$p(L+c) = \frac{-G}{\alpha} \tanh\left(\frac{\alpha L}{2}\right)$$

$\chi \neq 1$

$$\frac{p_x(L+c, y)}{p_x(c, y)} = \chi$$

$$p_x(L+c, y) = -\chi G$$